

LinkedIn Job Market Analytics

Phase 1 — Data Profiling, Cleaning, and Ingestion Report

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Big Data Application Development — Spring 2026

1. Project Context

This project investigates whether companies that experienced public layoffs between 2020 and 2026 hire differently than the rest of the US job market in 2023-2024. The analytical thesis is operationalised as a cohort comparison between layoff-affected and non-affected employers, evaluated across three angles: salary and experience-level mix, skills mix, and geographic and remote distribution.

Phase 1, the scope of this report, covers ingestion only: profiling and cleaning each raw data source and producing a reproducible cleaned dataset for downstream analytics. All work is performed in Apache Spark (PySpark) running inside the official jupyter/pyspark-notebook Docker image. Each script is invoked via spark-submit; no notebooks or interactive shells are used. Outputs are written as Snappy Parquet (for cleaned datasets) and single-file CSV (for human-readable profile summaries and run logs).

```
submission > Dhairya_dpm8739_phase1 > logs > 01_postings.log
1 26/04/24 21:07:17 INFO SparkContext: Running Spark version 3.5.0
2 26/04/24 21:07:17 INFO SparkContext: OS info Linux, 5.15.167.4-microsoft-standard-WSL2, amd64
3 26/04/24 21:07:17 INFO SparkContext: Java version 17.0.8.1
4 26/04/24 21:07:17 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
5 26/04/24 21:07:17 INFO ResourceUtils: =====
6 26/04/24 21:07:17 INFO ResourceUtils: No custom resources configured for spark.driver.
7 26/04/24 21:07:17 INFO ResourceUtils: =====
8 26/04/24 21:07:17 INFO SparkContext: Submitted application: 01_profile_and_clean_postings
9 26/04/24 21:07:17 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, amount: 1, script: , memo
10 26/04/24 21:07:17 INFO ResourceProfile: Limiting resource is cpu
11 26/04/24 21:07:17 INFO ResourceProfileManager: Added ResourceProfile id: 0
12 26/04/24 21:07:17 INFO SecurityManager: Changing view acls to: jovyan
13 26/04/24 21:07:17 INFO SecurityManager: Changing modify acls to: jovyan
14 26/04/24 21:07:17 INFO SecurityManager: Changing view acls groups to:
15 26/04/24 21:07:17 INFO SecurityManager: Changing modify acls groups to:
16 26/04/24 21:07:17 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view permissions: jovyan; groups with view
17 26/04/24 21:07:17 INFO Utils: Successfully started service 'sparkDriver' on port 40501.
18 26/04/24 21:07:17 INFO SparkEnv: Registering MapOutputTracker
19 26/04/24 21:07:17 INFO SparkEnv: Registering BlockManagerMaster
20 26/04/24 21:07:17 INFO BlockManagerMasterEndpoint: Using org.apache.spark.storage.DefaultTopologyMapper for getting topology information
21 26/04/24 21:07:17 INFO BlockManagerMasterEndpoint: BlockManagerMasterEndpoint up
22 26/04/24 21:07:17 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
23 26/04/24 21:07:17 INFO DiskBlockManager: Created local directory at /tmp/blockmgr-317c797b-b96c-42a9-b632-827b4b2b73d6
24 26/04/24 21:07:17 INFO MemoryStore: MemoryStore started with capacity 434.4 MiB
25 26/04/24 21:07:17 INFO SparkEnv: Registering OutputCommitCoordinator
26 26/04/24 21:07:17 INFO JettyUtils: Start Jetty 0.0.0.0:4040 for SparkUI
27 26/04/24 21:07:17 INFO Utils: Successfully started service 'SparkUI' on port 4040.
28 26/04/24 21:07:18 INFO Executor: Starting executor ID driver on host 8a6259461f28
29 26/04/24 21:07:18 INFO Executor: OS info Linux, 5.15.167.4-microsoft-standard-WSL2, amd64
30 26/04/24 21:07:18 INFO Executor: Java version 17.0.8.1
```

```
1087 26/04/24 21:10:39 INFO DAGScheduler: Got job 30 (count at NativeMethodAccessorImpl.java:0) with 1 output partitions
1088 26/04/24 21:10:39 INFO DAGScheduler: Final stage: ResultStage 49 (count at NativeMethodAccessorImpl.java:0)
1089 26/04/24 21:10:39 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 48)
1090 26/04/24 21:10:39 INFO DAGScheduler: Missing parents: List()
1091 26/04/24 21:10:39 INFO DAGScheduler: Submitting ResultStage 49 (MapPartitionsRDD[113] at count at NativeMethodAccessorImpl.java:0), which has no missing p.
1092 26/04/24 21:10:39 INFO MemoryStore: Block broadcast_45 stored as values in memory (estimated size 12.5 KiB, free 433.9 MiB)
1093 26/04/24 21:10:39 INFO MemoryStore: Block broadcast_45_piece0 stored as bytes in memory (estimated size 5.9 KiB, free 433.9 MiB)
1094 26/04/24 21:10:39 INFO BlockManagerInfo: Added broadcast_45_piece0 in memory on 8a6259461f28:39787 (size: 5.9 KiB, free: 434.3 MiB)
1095 26/04/24 21:10:39 INFO SparkContext: Created broadcast 45 from broadcast at DAGScheduler.scala:1580
1096 26/04/24 21:10:39 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 49 (MapPartitionsRDD[113] at count at NativeMethodAccessorImpl.java:0) (f.
1097 26/04/24 21:10:39 INFO TaskSchedulerImpl: Adding task set 49.0 with 1 tasks resource profile 0
1098 26/04/24 21:10:39 INFO TaskSetManager: Starting task 0.0 in stage 49.0 (TID 30) (8a6259461f28, executor driver, partition 0, NODE_LOCAL, 7615 bytes)
1099 26/04/24 21:10:39 INFO Executor: Running task 0.0 in stage 49.0 (TID 30)
1100 26/04/24 21:10:39 INFO ShuffleBlockFetcherIterator: Getting 1 (60.0 B) non-empty blocks including 1 (60.0 B) local and 0 (0.0 B) host-local and 0 (0.0 B)
1101 26/04/24 21:10:39 INFO ShuffleBlockFetcherIterator: Started 0 remote fetches in 0 ms
1102 26/04/24 21:10:39 INFO Executor: Finished task 0.0 in stage 49.0 (TID 30). 3952 bytes result sent to driver
1103 26/04/24 21:10:39 INFO TaskSetManager: Finished task 0.0 in stage 49.0 (TID 30) in 3 ms on 8a6259461f28 (executor driver) (1/1)
1104 26/04/24 21:10:39 INFO TaskSchedulerImpl: Removed TaskSet 49.0, whose tasks have all completed, from pool
1105 26/04/24 21:10:39 INFO DAGScheduler: ResultStage 49 (count at NativeMethodAccessorImpl.java:0) finished in 0.005 s
1106 26/04/24 21:10:39 INFO DAGScheduler: Job 30 is finished. Cancelling potential speculative or zombie tasks for this job
1107 26/04/24 21:10:39 INFO TaskSchedulerImpl: Killing all running tasks in stage 49: Stage finished
1108 26/04/24 21:10:39 INFO DAGScheduler: Job 30 finished: count at NativeMethodAccessorImpl.java:0, took 0.006751 s
1109 CLEAN: dropped[min_salary_gt_max_salary]=0
1110 26/04/24 21:10:39 INFO FileSourceStrategy: Pushed Filters:
1111 26/04/24 21:10:39 INFO FileSourceStrategy: Post-Scan Filters: ((isnull(min_salary#10) OR isnull(max_salary#4)) OR (cast(min_salary#10 as double) <= cast(m.
1112 26/04/24 21:10:39 INFO CodeGenerator: Code generated in 4.624807 ms
1113 26/04/24 21:10:39 INFO MemoryStore: Block broadcast_46 stored as values in memory (estimated size 199.8 KiB, free 433.7 MiB)
1114 26/04/24 21:10:39 INFO MemoryStore: Block broadcast_46_piece0 stored as bytes in memory (estimated size 34.3 KiB, free 433.6 MiB)
1115 26/04/24 21:10:39 INFO BlockManagerInfo: Added broadcast_46_piece0 in memory on 8a6259461f28:39787 (size: 34.3 KiB, free: 434.3 MiB)
1116 26/04/24 21:10:39 INFO SparkContext: Created broadcast 46 from count at NativeMethodAccessorImpl.java:0
```

```
1499 26/04/24 21:11:10 INFO CodecPool: Got brand-new compressor [.snappy]
1500 26/04/24 21:11:10 INFO FileScanRDD: Reading File path: file:///home/jovyan/work/data/linkedin/postings.csv, range: 0-516843769, partition values: [empty r
1501 26/04/24 21:11:10 INFO BlockManagerInfo: Removed broadcast_52_piece0 on 8a6259461f28:39787 in memory (size: 78.3 KiB, free: 434.1 MiB)
1502 26/04/24 21:11:10 INFO BlockManagerInfo: Removed broadcast_51_piece0 on 8a6259461f28:39787 in memory (size: 5.9 KiB, free: 434.1 MiB)
1503 26/04/24 21:11:30 INFO FileOutputCommitter: Saved output of task 'attempt_202604242111093209311084618011921_0057_m_000000_36' to file:/home/jovyan/work/ou
1504 26/04/24 21:11:30 INFO SparkHadoopMapRedUtil: attempt_202604242111093209311084618011921_0057_m_000000_36: Committed. Elapsed time: 24 ms.
1505 26/04/24 21:11:30 INFO Executor: Finished task 0.0 in stage 57.0 (TID 36). 2626 bytes result sent to driver
1506 26/04/24 21:11:30 INFO TaskSetManager: Finished task 0.0 in stage 57.0 (TID 36) in 20931 ms on 8a6259461f28 (executor driver) (1/1)
1507 26/04/24 21:11:30 INFO TaskSchedulerImpl: Removed TaskSet 57.0, whose tasks have all completed, from pool
1508 26/04/24 21:11:30 INFO DAGScheduler: ResultStage 57 (parquet at NativeMethodAccessorImpl.java:0) finished in 20.941 s
1509 26/04/24 21:11:30 INFO DAGScheduler: Job 36 is finished. Cancelling potential speculative or zombie tasks for this job
1510 26/04/24 21:11:30 INFO TaskSchedulerImpl: Killing all running tasks in stage 57: Stage finished
1511 26/04/24 21:11:30 INFO DAGScheduler: Job 36 finished: parquet at NativeMethodAccessorImpl.java:0, took 20.857883 s
1512 26/04/24 21:11:30 INFO FileFormatWriter: Start to commit write Job d2a84c86-4fec-47b1-9d59-fed37740c7b1.
1513 26/04/24 21:11:31 INFO FileFormatWriter: Write Job d2a84c86-4fec-47b1-9d59-fed37740c7b1 committed. Elapsed time: 213 ms.
1514 26/04/24 21:11:31 INFO FileFormatWriter: Finished processing stats for write job d2a84c86-4fec-47b1-9d59-fed37740c7b1.
1515 DONE: profile and parquet outputs written
1516 26/04/24 21:11:31 INFO SparkContext: SparkContext is stopping with exitCode 0.
1517 26/04/24 21:11:31 INFO SparkUI: Stopped Spark web UI at http://8a6259461f28:4040
1518 26/04/24 21:11:31 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
1519 26/04/24 21:11:31 INFO MemoryStore: MemoryStore cleared
1520 26/04/24 21:11:31 INFO BlockManager: BlockManager stopped
1521 26/04/24 21:11:31 INFO BlockManagerMaster: BlockManagerMaster stopped
1522 26/04/24 21:11:31 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
1523 26/04/24 21:11:31 INFO SparkContext: Successfully stopped SparkContext
1524 26/04/24 21:11:31 INFO ShutdownHookManager: Shutdown hook called
1525 26/04/24 21:11:31 INFO ShutdownHookManager: Deleting directory /tmp/spark-92d7eb6c-89b2-41c9-a91e-424e97dbdc04/pyspark-8a2ad8dc-339d-49c8-b732-67d3d545c78.
1526 26/04/24 21:11:31 INFO ShutdownHookManager: Deleting directory /tmp/spark-92d7eb6c-89b2-41c9-a91e-424e97dbdc04
1527 26/04/24 21:11:31 INFO ShutdownHookManager: Deleting directory /tmp/spark-ccb1ba7d-e4bb-4f77-8055-ada4675408bd
1528
```

[SCREENSHOTS: Terminal showing successful spark-submit run inside the Docker container]

2. Data Sources at a Glance

Three public datasets are ingested. As a solo team (professor-approved), all three are this submitter’s responsibility.

#	Source	Origin / License	Role in the analysis
1	LinkedIn Job Postings (2023–2024)	Kaggle: arshkon/linkedin-job-postings (~123k rows, ~500 MB CSV)	Primary observations: 2023–2024 US postings.
2	LinkedIn Skills mapping	Same Kaggle dataset; two files: jobs/job_skills.csv and mappings/skills.csv	Maps each job_id to a normalized list of skill tags.
3	Tech Layoffs (2020–2026)	Kaggle: swaptr/layoffs-2022 (continuously refreshed; ~4.3k rows)	Defines the layoff-affected cohort by company name.

3. Source 1 — LinkedIn Job Postings

The LinkedIn dataset contains ~123,849 raw job postings scraped during 2023-2024. The raw CSV has 31 columns and uses multi-line description fields with embedded quotes; all reads use header=True, multiLine=True, escape="'", quote="'" to parse correctly. Spark’s default CSV options misalign rows with embedded newlines.

Schema decisions

Eighteen columns are kept and thirteen are dropped. Kept columns: job_id, company_id, company_name, title, location, formatted_work_type, formatted_experience_level, remote_allowed, min_salary, max_salary, med_salary, pay_period, normalized_salary, currency, views, applies, original_listed_time, zip_code. Dropped columns: description, skills_desc, job_posting_url, application_url, posting_domain, application_type, closed_time, expiry, listed_time, sponsored,

work_type, compensation_type, fips. Dropped columns are either redundant (work_type vs formatted_work_type), high-volume free text (description, skills_desc), or PII/URL fields not relevant to the analysis.

Profiling findings (raw)

- Raw row count: 123,849.
- Null-rate highlights: company_name 1.4% (1,719 rows), formatted_experience_level 23.8% (29,409 rows), remote_allowed 87.7% (108,603 rows; treated as not-remote in cleaning), normalized_salary 71% (87,776 rows), zip_code 16.8% (20,872 rows).
- Distinct value counts: 7 work types, 7 experience levels, 7 currency codes (USD dominates).
- Numeric describe: normalized_salary mean approx \$94k, range from <\$1k (clear data errors) to >\$10M.

Cleaning rules applied (in order, with drop counts)

1. Project to the 18 kept columns (drops the 13 free-text/redundant/PII columns).
2. Split location on ',' to derive lower-cased, trimmed city and state columns; the original location column is retained for display.
3. Cast zip_code as StringType to preserve leading zeros (never cast to integer).
4. Convert original_listed_time (epoch milliseconds) to a new listed_date column via from_unixtime(col("original_listed_time")/1000).cast("date").
5. Drop rows where min_salary > max_salary when both are present (dropped 0 rows).
6. Drop rows where normalized_salary is outside [10,000, 1,000,000] sanity bounds; null normalized_salary rows are kept (dropped 455 rows).
7. Cast remote_allowed to Boolean: 1 → true, anything else (0, null, empty) → false.
8. Add company_normalized via lower(trim(regex_replace(company_name, "(?i)\\s*(inc|corp|corporation|llc|Ltd|limited)\\.?\$", ""))) for downstream joins.

Final dataset

Cleaned row count: 123,394 (455 rows dropped, ~0.37% of raw). Output: output/parquet/postings/ as Snappy Parquet, ~6.1 MB. Profile summary: output/profiles/postings/ in long form (dataset, column, metric, value). Sample: snippets/postings_sample.csv (20 rows, 22 columns).

submission > Dhairya_dpm8739_phase1 > snippets >  postings_sample.csv

	job_id	company_id	company_name	title	location	formatted_work_type	fo
1	921716	2774458.0	Corcoran Sawyer Smith	Marketing Coordinator	Princeton, NJ	Full-time	
2	1829192			Mental Health Therapist/Counselor	Fort Collins, CO	Full-time	
3	10998357	64896719.0	The National Exemplar	Assitant Restaurant Manager	Cincinnati, OH	Full-time	
4	23221523	766262.0	Abrams Fensterman, LLP	Senior Elder Law / Trusts and Estates Associate Attorney	New Hyde Park, NY	Full-time	
5	35982263			Service Technician	Burlington, IA	Full-time	
6	91700727	1481176.0	Downtown Raleigh Alliance	Economic Development and Planning Intern	Raleigh, NC	Internship	
7	103254301	81942316.0	Raw Cereal	Producer	United States	Contract	
8	112576855			Building Engineer	San Francisco, CA	Full-time	
9	1218575	721189.0	Children's Nebraska	Respiratory Therapist	Omaha, NE	Full-time	
10	9615617	1194336.0	Glastender, Inc.	Inside Customer Service Associate	Saginaw, MI	Full-time	
11	11009123	93456.0	PGAV Destinations	Project Architect	St Louis, MO	Full-time	
12	56482768			Appalachian Highlands Women's Business Center	Kingsport, TN	Full-time	
13	56924323			Structural Engineer	San Mateo, CA	Full-time	
14	69333422	27097933.0	Staffing Theory	Senior Product Marketing Manager	United States	Full-time	
15	83789755	3810432.0	OsteoStrong	Osteogenic Loading Coach	Anchorage, AK	Full-time	
16	95428182	55624331.0	CLEVELAND KIDS BOOK BANK	Administrative Coordinator	Cleveland, OH	Full-time	
17	103860943			Customer Service / Reservationist	Providence, RI	Part-time	
18	111513530	4028816.0	United Methodists of Greater New Jersey	Content Writer, Communications	Greater Philadelphia	Full-time	
19	115639136	988555.0	Shannon Waltchack	Controller	Birmingham, AL	Full-time	
20	117675818	4285947.0	Premier Family Clinic	Physician Assistant	Atlanta, GA	Contract	
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remote_allowed	min_salary	max_salary	med_salary	pay_period	normalized_salary	currency	views	applies	original_listed_time	zip_code	city	state	listed
false	17.0	20.0		HOURLY	38480.0	USD	20.0	2.0	1713397508000.0	08540	princeton	nj	2024-0
false	30.0	50.0		HOURLY	83200.0	USD	1.0		1712857887000.0	80521	fort collins	co	2024-0
false	45000.0	65000.0		YEARLY	55000.0	USD	8.0		1713277614000.0	45202	cincinnati	oh	2024-0
false	140000.0	175000.0		YEARLY	157500.0	USD	16.0		1712895812000.0	11040	new hyde park	ny	2024-0
false	60000.0	80000.0		YEARLY	70000.0	USD	3.0		1713451943000.0	52601	burlington	ia	2024-0
false	14.0	20.0		HOURLY	35360.0	USD	9.0	4.0	1713456099000.0	27601	raleigh	nc	2024-0
true	60000.0	300000.0		YEARLY	180000.0	USD	7.0	1.0	1712861019000.0		united states		2024-0
false	90000.0	120000.0		YEARLY	105000.0	USD	2.0		1712443452000.0	94101	san francisco	ca	2024-0
false							3.0		1712348500000.0	68102	omaha	ne	2024-0
false							4.0	1.0	1713210566000.0	48601	saginaw	mi	2024-0
false							2.0		1712858322000.0	63367	st louis	mo	2024-0
false							2.0		1712870267000.0	37660	kingsport	tn	2024-0
false							2.0		1712460580000.0	94401	san mateo	ca	2024-0
true							2.0		1713537397000.0		united states		2024-0
false							4.0		1713466434000.0	99501	anchorage	ak	2024-0
false			25.0	HOURLY	52000.0	USD	1.0		1712855686000.0	44102	cleveland	oh	2024-0
false	18.0	19.0		HOURLY	38480.0	USD	3.0		1713532302000.0	02903	providence	ri	2024-0
false							10.0	1.0	1713457320000.0		greater philadelphia		2024-0
false							61.0	4.0	1713209850000.0	35203	birmingham	al	2024-0
false							29.0	1.0	1712667107000.0	30303	atlanta	ga	2024-0

submission > Dhairya_dpm8739_phase1 > snippets >  layoffs_sample.csv

	company	location	total_laid_off	date	percentage_laid_off	industry	stage	country	company_normalized
1	Shutterfly	Haifa, Non-U.S.	80	2026-04-16		Manufacturing	Post-IPO	United States	shutterfly
2	Snap	Los Angeles	1000	2026-04-15	0.16	Consumer	Post-IPO	United States	snap
3	UKG	Boston	950	2026-04-15		HR	Unknown	United States	ukg
4	Productboard	SF Bay Area		2026-04-15	0.3	Product	Series D	United States	productboard
5	Cars.com	Chicago		2026-04-09	0.11	Transportation	Acquired	United States	cars.com
6	Pepper Pay	Miami		2026-04-09	1.0	Finance	Unknown	United States	pepper pay
7	GoPro	SF Bay Area	145	2026-04-07	0.23	Consumer	Post-IPO	United States	gopro
8	Pendo	Raleigh	90	2026-04-07	0.1	Product	Series F	United States	pendo
9	Bolt	SF Bay Area		2026-04-05	0.33	Finance	Series E	United States	bolt
10	NeuroPixel.AI	Bengaluru, Non-U.S.		2026-04-03	1.0	Retail	Seed	India	neuropixel.ai
11	Meta	SF Bay Area	200	2026-04-02		Consumer	Post-IPO	United States	meta
12	Oracle	SF Bay Area	30000	2026-03-31		Other	Post-IPO	United States	oracle
13	Monzo	London, Non-U.S.	50	2026-03-31		Finance	Series F	United Kingdom	monzo
14	Covrzy	Bengaluru, Non-U.S.		2026-03-31	1.0	Finance	Seed	India	covrzy
15	Rec Room	Seattle		2026-03-31	1.0	Consumer	Series F	United States	rec room
16	Yupp	SF Bay Area		2026-03-31	1.0	AI	Seed	United States	yupp
17	Enpal	Berlin, Non-U.S.	100	2026-03-27		Energy	Series E	Germany	enpal
18	Monte Carlo	SF Bay Area		2026-03-26	0.3	Data	Series D	United States	monte carlo
19	Quandoo	Berlin, Non-U.S.		2026-03-26	1.0	Food	Acquired	Germany	quandoo
20	Epic Games	Raleigh	1000	2026-03-24		Consumer	Unknown	United States	epic games
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[SCREENSHOT: Sample rows from postings_sample.csv and layoffs_sample.csv]

4. Source 2 — LinkedIn Skills Mapping

Two files: jobs/job_skills.csv (job_id ↔ skill_abr, ~213,768 rows) and mappings/skills.csv (skill_abr → skill_name, 35 distinct skill codes such as ENG, IT, ANLS, MGMT). The mapping table is small and well-defined; the join is one-to-many on the skill code.

Cleaning rules applied (in order, with drop counts)

- 1. Profile both files independently (schema, row count, null counts, distinct counts on skill_abr).
- 2. Left join job_skills to skills on skill_abr.
- 3. Drop rows where skill_name is null after the join (orphan skill codes; dropped 0 rows).
- 4. Deduplicate on (job_id, skill_abr) (dropped 0 rows).

Final dataset

Cleaned row count: 213,768 with three columns (job_id, skill_abr, skill_name). No orphans and no duplicates were present in the source; the join is integrity-preserving. Output: output/parquet/skills/ as Snappy Parquet, ~1.4 MB. Sample: snippets/skills_sample.csv.

5. Source 3 — Tech Layoffs (2020–2026)

Continuously refreshed Kaggle dataset. The label “layoffs-2022” in the slug is historical; the file currently spans 2020-03-11 to 2026-04-16 (4,357 raw rows). Date strings use the M/d/yyyy format. Some rows have null total_laid_off but a non-null percentage_laid_off, and vice versa.

Schema decisions

Kept columns: company, location, total_laid_off, date, percentage_laid_off, industry, stage, country. Dropped columns: source, funds_raised, date_added (irrelevant or downstream metadata).

Cleaning rules applied (in order, with drop counts)

1. Parse date with format M/d/yyyy into a DateType column (replace the original).
2. Drop rows where BOTH total_laid_off AND percentage_laid_off are null (dropped 714 rows, 16.4%).
3. Cast total_laid_off as IntegerType.
4. Cast percentage_laid_off as DoubleType.
5. Add company_normalized using the same regex as the postings source so the two datasets can be joined on a normalized key.

Final dataset

Cleaned row count: 3,643. Output: output/parquet/layoffs/ as Snappy Parquet, ~89 KB. Sample: snippets/layoffs_sample.csv (20 rows, 9 columns).

6. Spark Techniques Used

All three ingestion scripts use the high-level DataFrame API together with Spark SQL functions. RDDs are not used directly; everything is expressed as DataFrame transformations and aggregations. The full list of Spark constructs used across scripts 01–03 is summarized below.

- Cluster + session: SparkSession.builder.appName(...).getOrCreate(); spark.stop() at the end of each script for clean shutdown.
- Ingestion: spark.read.option("header", True).option("multiLine", True).option("escape", "").option("quote", "").csv(path) for quote-embedded CSVs.
- Profiling: df.printSchema(), df.count(), df.distinct().count(), df.describe(), df.show(n, truncate=False) for sample rows.
- Null profiling: a single aggregation that computes per-column null counts via F.sum(F.when(F.col(c).isNull(), 1).otherwise(0)).alias(c) over all columns.
- Top-N value counts on key categoricals via groupBy(col).count().orderBy(F.desc("count")).show(10).
- Transformations: select, withColumn, filter, dropDuplicates(["job_id", "skill_abr"]), join (left).
- Spark SQL functions used: F.col, F.lit, F.when, F.lower, F.trim, F.regexp_replace, F.split, F.from_unixtime, F.to_date, F.sum, F.desc.
- Type casting: .cast("int"), .cast("double"), .cast("string"), .cast("date") for explicit type control after ingestion.

- Persistence: `df.write.mode("overwrite").parquet(path)` for cleaned datasets;
`df.coalesce(1).write.mode("overwrite").option("header", True).csv(path)` for human-readable single-file profile CSVs.
- Long-form profile output construction: `spark.createDataFrame(metrics_list, schema=[...])` builds a (dataset, column, metric, value) DataFrame from an in-memory list of tuples, providing a uniform shape for all three sources.

Logging is grep-friendly: each script prints labelled lines such as PROFILE:, CLEAN:, and DONE: to stdout so the run log itself is the audit trail for what happened during ingestion.

7. Data Quality Notes and Limitations

- Postings: approximately 1.4% of cleaned rows (1,709) have null `company_normalized`. These rows are kept in the cleaned dataset but will be excluded from any cohort analysis that requires the company key.
- Postings: ~71% of rows have null `normalized_salary` in raw form. Salary-based queries downstream will filter to non-null USD rows; this is documented as a deferred query-time filter rather than a cleaning-time drop.
- Layoffs: dataset slug labels it “2022” but the data is continuously updated through 2026; current cut spans 2020-03-11 to 2026-04-16.
- Skills: zero orphans and zero duplicates in the raw join; this is convenient and likely reflects Kaggle’s upstream curation rather than a robust property of LinkedIn data in general.
- All three sources are static snapshots/refreshes. The project does not yet integrate streaming or live data.

8. How to Run

From the project root, in PowerShell:

```
docker run --rm -v "${PWD}:/home/jovyan/work" --memory=8g jupyter/pyspark-notebook bash -lc "set -e; cd /home/jovyan/work && spark-submit scripts/01_profile_and_clean_postings.py 2>&1 | tee output/logs/01_postings.log && spark-submit scripts/02_profile_and_clean_skills.py 2>&1 | tee output/logs/02_skills.log && spark-submit scripts/03_profile_and_clean_layoffs.py 2>&1 | tee output/logs/03_layoffs.log"
```

Total runtime: approximately 5 minutes on a laptop with 8 GB RAM allocated to Docker. Outputs land under `output/parquet/`, `output/profiles/`, and `output/logs/`.

```
PS C:\--DPM-MAIN-DIR--\--GIT-REPOS--\job-posting-analytics-spark> docker run --rm -v "${PWD}:/home/jovyan/work" --memory=8g jupyter/pyspark-notebook bash -lc "set -e; cd /home/jovyan/work && spark-submit scripts/01_profile_and_clean_postings.py 2>&1 | tee output/logs/01_postings.log && spark-submit scripts/02_profile_and_clean_skills.py 2>&1 | tee output/logs/02_skills.log && spark-submit scripts/03_profile_and_clean_layoffs.py 2>&1 | tee output/logs/03_layoffs.log"
26/04/25 02:23:55 INFO SparkContext: Running Spark version 3.5.0
26/04/25 02:23:55 INFO SparkContext: OS info Linux, 5.15.167.4-microsoft-standard-WSL2, amd64
26/04/25 02:23:55 INFO SparkContext: Java version 17.0.8.1
26/04/25 02:23:55 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
26/04/25 02:23:55 INFO ResourceUtils: =====
26/04/25 02:23:55 INFO ResourceUtils: No custom resources configured for spark.driver.
26/04/25 02:23:55 INFO ResourceUtils: =====
26/04/25 02:23:55 INFO SparkContext: Submitted application: 01_profile_and_clean_postings
26/04/25 02:23:55 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, amount: 1, script: , vendor: , memory -> name: memory, amount: 1024, script: , vendor: , offHeap -> name: offHeap, amount: 0, script: , vendor: ), task resources: Map(cpus -> name: cpus, amount: 1.0)
26/04/25 02:23:55 INFO ResourceProfile: Limiting resource is cpu
26/04/25 02:23:55 INFO ResourceProfileManager: Added ResourceProfile id: 0
26/04/25 02:23:55 INFO SecurityManager: Changing view acls to: jovyan
26/04/25 02:23:55 INFO SecurityManager: Changing modify acls to: jovyan
26/04/25 02:23:55 INFO SecurityManager: Changing view acls groups to:
26/04/25 02:23:55 INFO SecurityManager: Changing modify acls groups to:
26/04/25 02:23:55 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view permissions: jovyan; groups with view permissions: EMPTY; users with modify permissions: jovyan; groups with modify permissions: EMPTY
26/04/25 02:23:55 INFO Utils: Successfully started service 'sparkDriver' on port 38347.
26/04/25 02:23:55 INFO SparkEnv: Registering MapOutputTracker
26/04/25 02:23:55 INFO SparkEnv: Registering BlockManagerMaster
```

```
Containers / naughty_hofstadter
naughty_hofstadter
@ 25b88294e4ba @ juvyter/zyospark.notebook.latest
STATUS
Running (55 seconds ago)

Logs Inspect Bind mounts Exec Files Stats

26/04/25 02:23:55 INFO SparkContext: Running Spark version 3.5.0
26/04/25 02:23:55 INFO SparkContext: OS info Linux, 5.15.167.4-microsoft-standard-WSL2, amd64
26/04/25 02:23:55 INFO SparkContext: Java version 17.0.8.1
26/04/25 02:23:55 WARN NativeCodeLoader: Unable to load native-heap library for your platform... using builtin-java classes where applicable
26/04/25 02:23:55 INFO ResourceUtils: =====
26/04/25 02:23:55 INFO ResourceUtils: No custom resources configured for spark driver.
26/04/25 02:23:55 INFO ResourceUtils: =====
26/04/25 02:23:55 INFO SparkContext: Submitted application: 01_profile_and_clean_postings
26/04/25 02:23:55 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, amount: 1, script: , vendor: , memory -> name: memory, amount: 1024, script: , vendor: , offheap -> name: offheap, amount: 0, script: , vendor: ), task resources: Map(cpus -> name: cpus, amount: 1.0)
26/04/25 02:23:55 INFO ResourceProfile: Limiting resource is cpu
26/04/25 02:23:55 INFO ResourceProfileManager: Added ResourceProfile id: 0
26/04/25 02:23:55 INFO SecurityManager: Changing view acls to: jovyana
26/04/25 02:23:55 INFO SecurityManager: Changing modify acls to: jovyana
26/04/25 02:23:55 INFO SecurityManager: Changing view acls groups to:
26/04/25 02:23:55 INFO SecurityManager: Changing modify acls groups to:
26/04/25 02:23:55 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view permissions: jovyana; groups with view permissions: EMPTV; users with modify permissions: jovyana; groups with modify permissions: EMPTV
26/04/25 02:23:55 INFO Utils: Successfully started service 'sparkDriver' on port 38347.
26/04/25 02:23:55 INFO SparkEnv: Registering MapOutputTracker
26/04/25 02:23:55 INFO SparkEnv: Registering BlockManagerMaster
26/04/25 02:23:55 INFO BlockManagerMasterEndpoint: Using org.apache.spark.storage.DefaultTopologyMapper for getting topology information
26/04/25 02:23:55 INFO BlockManagerMasterEndpoint: BlockManagerMasterEndpoint up
26/04/25 02:23:55 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
26/04/25 02:23:56 INFO DiskBlockManager: Created local directory at /tmp/blockmgr-2b51e2d4-88be-49c3-8225-6ff7fac7a77
26/04/25 02:23:56 INFO MemoryStore: MemoryStore started with capacity 434.4 MiB
26/04/25 02:23:56 INFO SparkEnv: Registering OutputCommitCoordinator
26/04/25 02:23:56 INFO JettyUtils: Start Jetty 9.4.44.v20210824 for SparkUI
26/04/25 02:23:56 INFO Utils: Successfully started service 'SparkUI' on port 4040.
26/04/25 02:23:56 INFO Executor: Starting executor ID driver on host 25b88294e4ba
26/04/25 02:23:56 INFO Executor: OS info Linux, 5.15.167.4-microsoft-standard-WSL2, amd64
26/04/25 02:23:56 INFO Executor: Java version 17.0.8.1
26/04/25 02:23:56 INFO Executor: Starting executor with user classpath (userClassPathFirst = false): ''
26/04/25 02:23:56 INFO Executor: Created or updated repl class loader org.apache.spark.util.MutableURLClassLoader@4c171869 for default.
```

[SCREENSHOTS: Terminal showing the three scripts completing successfully end-to-end]

9. Submission File Inventory

- scripts/ — three .py ingestion scripts (01, 02, 03), each runnable standalone via spark-submit.
- profiles/ — three single-file CSVs (one per source) in long form (dataset, column, metric, value).
- logs/ — three full stdout logs from a successful spark-submit run, with grep-friendly labels.
- snippets/ — three 20-row CSVs sampled from the cleaned Parquet datasets to illustrate final shape.